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|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| EDUCATION             | <b>The Australian National University (ANU)</b><br><i>Ph.D. in Computer Science</i><br>• Advisor: Dr. <a href="#">Piotr Koniusz</a> , Dr. <a href="#">Liang Zheng</a> , Prof. <a href="#">Stephen Gould</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Canberra, Australia<br>2020.10 - 2025.01 |
|                       | <b>Shanghai Jiao Tong University (SJTU)</b><br><i>M.S. in Pattern Recognition and Intelligent System</i><br>• Advisor: Prof. <a href="#">Xinping Guan</a> (IEEE Fellow)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Shanghai, China<br>2017.09 - 2020.03     |
|                       | <b>Southeast University (SEU), Department of Automation</b><br><i>B.E. in Automation</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Nanjing, China<br>2013.09 - 2017.06      |
| WORK EXPERIENCE       | <b>Hong Kong University of Science and Technology (HKUST)</b><br><i>Postdoc Fellow, Research Assistant Professor</i><br>• Advisor: Prof. <a href="#">Jiaya Jia</a> (IEEE Fellow)<br>• Investigating multi-modal generic vision, VLM and VLA; Writing grant proposals; Teaching and supervising students                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Hong Kong<br>2025.07 - 2026.08 - Now     |
|                       | <b>Australian Institute for Machine Learning, Adelaide University</b><br><i>Research Fellow</i><br>• Advisor: Prof. <a href="#">Anton van den Hengel</a> (Fellow of Australian Academy of Tech. and Eng.)<br>• Investigating efficient and compact ML architectures and multi-modal generic vision                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Adelaide, Australia<br>2025.04 - 2025.07 |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                          |
| RESEARCH INTEREST     | I have wide research interests in <b>computer vision, machine learning, AI, and robotics</b> . My long-term pursuit is <b>artificial general intelligence</b> , <i>i.e.</i> , making the machine perceive, think and conduct more like a human.<br>Now I focus on <b>multimodal generic vision, foundation model, embodied intelligence (VLA, world model), zero- and few-shot learning, and transfer learning</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                          |
| SELECTED PUBLICATIONS | * indicates equal contribution; † indicates corresponding author                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                          |
|                       | <ol style="list-style-type: none"><li>1. <b>Changsheng Lu</b>, Yuxin Chen, Haokun Gui, Rong Wang, Jie Yang, Harry Yang, Anton van den Hengel, Jiaya Jia. “GKDT: General Keypoint Detection Transformer.” In <i>19th European Conference on Computer Vision (ECCV)</i>, 2026. (<a href="#">code</a>, <a href="#">dataset</a> &amp; <a href="#">demo</a>)</li><li>2. <b>Changsheng Lu</b>, Hao Zhu, and Piotr Koniusz. Exploiting Class-agnostic Visual Prior for Few-shot Keypoint Detection. <i>International Journal of Computer Vision (IJCV)</i>, 2025. (<b>Impact Factor: 19.5</b>)</li><li>3. Jiawei Cao, Chaochen Gu, Hao Cheng, Xiaofeng Zhang, Kaijie Wu<sup>†</sup>, <b>Changsheng Lu</b><sup>†</sup>. “EFDTR: Learnable Elliptical Fourier Descriptor Transformer for Instance Segmentation.” In <i>Proceedings of 42nd International Conference on Machine Learning (ICML)</i>, 2025.</li><li>4. <b>Changsheng Lu</b>, Zheyuan Liu, Piotr Koniusz. “OpenKD: Opening Prompt Diversity for Zero- and Few-Shot Keypoint Detection.” In <i>18th European Conference on Computer Vision (ECCV)</i>, 2024. (<a href="#">code</a>)</li><li>5. <b>Changsheng Lu</b>, Piotr Koniusz. “Detect Any Keypoints: An Efficient Light-weight Few-shot Keypoint Detector.” In <i>Proceedings of 38th Annual AAAI Conference on Artificial Intelligence (AAAI)</i>, 2024.</li><li>6. <b>Changsheng Lu</b>, Piotr Koniusz. “Few-shot Keypoint Detection with Uncertainty Learning for Unseen Species.” In <i>IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)</i>, 2022. (<a href="#">code</a>)</li><li>7. <b>Changsheng Lu</b>, Siyu Xia, Ming Shao, and Yun Fu. Arc-support Line Segments Revisited: An Efficient and High-quality Ellipse Detection. <i>IEEE Transactions on Image Processing (TIP)</i>, vol. 29, pp. 768-781, 2020, doi: 10.1109/TIP.2019.2934352. (<b>Impact Factor: 10.8</b>, <a href="#">code</a>)</li></ol> |                                          |

8. **Changsheng Lu\***, Wenlong Shi\*, Ming Shao, Yinjie Zhang, Siyu Xia, and Piotr Koniusz. “Few-shot Shape Recognition by Learning Deep Shape-aware Features.” In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2024.
9. **Changsheng Lu**, Chaochen Gu, Kaijie Wu, Siyu Xia, Haotian Wang, Xinping Guan. Deep transfer neural network using hybrid representations of domain discrepancy. *Neurocomputing*, 2020. (**Impact Factor: 6.0**)
10. **Changsheng Lu**, Haotian Wang, Chaochen Gu, Kaijie Wu, and Xinping Guan. “Viewpoint Estimation for Workpieces with Deep Transfer Learning from Cold to Hot.” In *International Conference on Neural Information Processing*, pp. 21-32. Springer, Cham, 2018. (**Oral, code**)
11. **Changsheng Lu**, Siyu Xia, Wanming Huang, Ming Shao, and Yun Fu. “Circle Detection by Arc-support Line Segments.” In *IEEE International Conference on Image Processing (ICIP)*, 2017. (**Oral, code**)
12. Dengyang Jiang, Dongyang Liu, Zanyi Wang, Qilong Wu, Liuzhuozheng Li, Hengzhuang Li, Xin Jin, David Liu, **Changsheng Lu**, Zhen Li, Bo Zhang, Mengmeng Wang, Steven Hoi, et al. “Distribution matching distillation meets reinforcement learning.” In *19th European Conference on Computer Vision (ECCV)*, 2026. (**code**)
13. Yexin Liu, Manyuan Zhang, Yueze Wang, Hongyu Li, Dian Zheng, Weiming Zhang, **Changsheng Lu**, Xunliang Cai, Yan Feng, Peng Pei, Harry Yang. “OpenSubject: Leveraging Video-Derived Identity and Diversity Priors for Subject-driven Image Generation and Manipulation.” In *19th European Conference on Computer Vision (ECCV)*, 2026. (**code**)
14. Haokun Gui, Senqiao Yang, Mingkan Zhu, Meng Chu, Sitong Wu, **Changsheng Lu**, Zihao Wang, Zhuotao Tian, Jiaya Jia. “VisionLeaf: Entropy-Guided Leaf-First Reasoning for Efficient and Accurate Think-with-Image.” In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.
15. Rong Wang, Wei Mao, **Changsheng Lu**, and Hongdong Li. “Towards High-Quality 3D Motion Transfer for Stylized Characters with Realistic Apparel Animation.” In *18th European Conference on Computer Vision (ECCV)*, 2024. (**code**)
16. Tongkun Guan, Chaochen Gu, **Changsheng Lu**, Jingzheng Tu, Qi Feng, Kaijie Wu, Xinping Guan. Industrial Scene Text Detection with Refined Feature-attentive Network. *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, 2022. (**Impact Factor: 8.4, code**)

## REPRESENTATIVE RESEARCH

- **Few-shot keypoint detection:** Keypoint detection is a fundamental task in computer vision. To break the limitation of keypoint types to be detected, we are the first to expand few-shot learning to the field of keypoint detection and open the door to this new field. Now there are over 6 research groups globally working on this new topic.
- **General keypoint detection:** To overcome the insufficient prompt diversity in modality, semantics (seen vs. unseen), and language, we propose a general keypoint detector which can perform both zero-shot and few-shot keypoint detection by exploiting the foundation models such as **CLIP** and **LLM** (e.g. GPT3.5, Vicuna). Further, we present a state-of-the-art **GKDT** model and large-scale **MegaKPT** dataset to enable the model with high practicality in real-world applications.
- **High-quality circle/ellipse detector:** Circle/ellipse detection are fundamental problems in digital image processing. We propose an industry-level circle/ellipse detection algorithm which can detect circles/ellipses from digital images precisely, robustly, and fast, overcoming the long-standing issue of unsatisfactory detection in this field. This research has gained high recognition from the global experts in computer vision, robotics, and material science, *etc.* (Rank #1 in [GitHub](#)).

## SELECTED AWARDS AND HONORS

### Awards:

- |                                                                                             |      |
|---------------------------------------------------------------------------------------------|------|
| • <b>Outstanding award</b> for studying abroad, awarded by CSC                              | 2023 |
| • <b>Best thesis award</b> for masters, by Shanghai Association of Automation (SAA)         | 2021 |
| • <b>AGRTP PhD Fellowship</b> , awarded by ANU & Australian Government                      | 2020 |
| • <b>National Scholarship</b> , awarded by Ministry of Education of China ( <b>top 2%</b> ) | 2019 |

- **National Scholarship**, awarded by Ministry of Education of China (**top 2%**) 2017
- **First-class thesis award** for undergraduates (**top 1%**) 2017
- **Envision Future Scholarship**, awarded by Envision Energy 2017
- **Top Ten Presentation** of the 7th student's academic presentation of Southeast University 2017
- **Principal's Scholarship**, awarded by Southeast University (**top 2%**) 2014

#### Honors:

- **Outstanding Reviewer for ECCV'22** 2022
- **Outstanding Master graduate** of SJTU in Shanghai (**top 1%**) 2020
- **Outstanding Undergraduate** of SEU 2017

#### Competition Awards:

- **First Prize of National Electronic Design Competition** 2016
- **First Prize of National Electronic Design Competition** 2015
- **First Prize in Advanced Mathematics Competition** 2014

## ACADEMIC SERVICES

### Journal Services

- IEEE Trans. on Pattern Analysis and Machine Intelligence (TPAMI)
- International Journal of Computer Vision (IJCV)
- IEEE Transactions on Image Processing (IEEE T-IP)
- IEEE Trans. on Circuits and Systems for Video Technology (TCSVT)
- Pattern Recognition (PR)
- Neural Processing Letters (NPL)
- IEEE/CAA Journal of Automatica Sinica
- IEEE Robotics and Automation Letters (RA-L)

### Conference Services

- CVPR, ICCV, ECCV
- AAAI, NeurIPS, ICML, ICLR
- WACV, BMVC, ICIP, IJCNN, ACCV, ICONIP, ICMLA, ICASSP

## TALKS

- 2024.10.03 "OpenKD: Opening Prompt Diversity for Zero- and Few-shot Keypoint Detection", *ECCV 2024*, Milan, Italy
- 2024.08.08 "General Keypoint Detection: Few-shot, Zero-shot and Beyond", *AI, ML and Friends Seminars, Australian National University (ANU)*, Australia
- 2024.04.17 "Few-Shot Keypoint Detection", *statML reading group, DATA61 / CSIRO*, Australia
- 2024.01.12 "Detect Any Keypoints: An Efficient Light-weight Few-shot Keypoint Detector", *AAAI 2024*
- 2022.06.24 "Few-shot Keypoint Detection with Uncertainty Learning for Unseen Species", *CVPR 2022*
- 2022.04.07 "Ellipse Detection: A Perspective from Low-level Vision to Deep Learning", *Dalian University of Technology (DLUT)*
- 2020.07.08 "High-precision steel ball surface defect detection", *Shanghai Steel Ball Plant Co. Ltd.*
- 2020.04.26 "How to do research?", *Southeast University (SEU)*
- 2020.01.13 "Research on Technologies of Transfer Learning towards Virtual-real Viewpoint Estimation", *Shanghai Jiao Tong Uni. (SJTU)*
- 2019.10.24 "Brief Introduction of AI Hotspots and My works", *Shanghai Jiao Tong Uni. (SJTU)*
- 2019.07.14 "PointDoN: A shape pattern aggregation module for deep learning on point cloud", *IJCNN 2019*, Budapest, Hungary
- 2018.12.13 "Viewpoint estimation for workpieces with deep transfer learning from cold to hot", *ICONIP 2018*, Siem Reap, Cambodia
- 2018.06.27 "SNc Neuron Detection Method Based on Deep Learning for Efficacy Evaluation of Anti-PD Drugs", *2018 American Control Conference*, Milwaukee, USA

## PROFESSIONAL EXPERIENCE

### Queensland University of Technology

Brisbane

*Remote Research Assistant*

2024.10 - 2024.12

- Participating the research on multi-modal pre-training for 3D vision understanding.

### Australian National University

Canberra

*PhD Candidate*

2020.10 - 2024.09

- Studying and developing novel algorithms for few-shot, one-shot, and zero-shot keypoint detection; Participating the research of shape recognition, motion transfer, *etc.*

### Shanghai Jiao Tong University

Shanghai

*Research Lead*

2020.07 - 2022.10

- Advising students to do research; Developing algorithms for COVID-19 test bands detection and nanometer thin membrane defect detection.

### Jiuding Automation Co. Ltd.

Nanjing

*Research Lead*

2020.03 - 2020.06

- Studying and developing high-precision steel ball surface defect detection system.

### Shanghai Jiao Tong University

Shanghai

*M.S. candidate*

2017.09 - 2020.03

- Studying and developing deep transfer learning methods for virtual-real workpiece viewpoint estimation; Participating the research of breast lesion segmentation, 3D point cloud processing, and generic vision recognition such as classification, detection and segmentation.

### Joint Stars Technology Co. Ltd

Nanjing

*Research Intern*

2016.10 - 2017.05

- Developing industry-level defect inspection algorithms. During this period, I learned lots of skills and knowledge regarding industrial standards and engineering.

### Huawei Nanjing Research Institute

Nanjing

*Engineer Intern*

2016 Summer

- Participating the embedded communication software programming (C/C++); I was listed as a member in the first term of Huawei F(X) future scientist program and achieved the special offer.

### Southeast University

Nanjing

*B.E. candidate*

2013.09 - 2017.06

- Studying advanced mathematics, control theory, digital signal processing, *etc.*; Establishing lots of interesting projects, including *software projects*: Tetrix, TSP (Traveling Salesman Problem), android travel software, *etc.*, and *automatic controlled systems*: smart car system, inverted/wind pendulum control system, and laser targeted shooting system; Cultivating broad interests ranging from the hardware to high-level algorithms.

## TEACHING EXPERIENCE

### Hong Kong University of Science and Technology (HKUST)

*Teaching Assistant & Administer (work with Prof. Jiaya Jia)*

- Build Your First AI Starup (*COMP6211K*) 2026.07 - 2026.08
- Build Your First AI Starup (*COMP6211K*) 2025.07 - 2025.08

### Shanghai Jiao Tong University (SJTU)

*Teaching Assistant*

- Advanced Academic Writing 2018.09 - 2019.01
- C++ Programming 2018.01 - 2018.06

## SUPERVISION

### PhD students

|             |              |                                      |
|-------------|--------------|--------------------------------------|
| Yuxin Chen  | 2025-Present | w./ Prof. Jiaya Jia, HKUST (ECCV'26) |
| Zixuan Wang | 2025-Present | w./ Prof. Jiaya Jia, HKUST           |
| Haoze Zheng | 2025-Present | w./ Prof. Harry Yang, HKUST          |

|            |           |                                                       |
|------------|-----------|-------------------------------------------------------|
| Rong Wang  | 2022-2026 | w./ Prof. Hongdong Li, ANU ( <b>ECCV'24, 3DV'26</b> ) |
| Jiawei Cao | 2023-2025 | w./ Prof. Kaijie Wu, SJTU ( <b>ICML'25</b> )          |

#### Master students

|              |           |                                                  |
|--------------|-----------|--------------------------------------------------|
| Wenlong Shi  | 2021-2024 | w./ Prof. Siyu Xia, SEU ( <b>WACV'24</b> )       |
| Tianhao Wang | 2020-2023 | w./ Prof. Siyu Xia, SEU ( <b>ACCV'22</b> )       |
| Jiawei Cao   | 2020-2023 | w./ Prof. Kaijie Wu, SJTU ( <b>ICONIP'22</b> )   |
| Tongkun Guan | 2020-2023 | w./ Prof. Chaochen Gu, SJTU ( <b>TCSVT'22</b> )  |
| Xunjin Wu    | 2019-2022 | w./ Prof. Chaochen Gu, SJTU ( <b>ICCAIS'21</b> ) |

#### Bachelor students

|              |           |                                                    |
|--------------|-----------|----------------------------------------------------|
| Yuanqi He    | 2020-2021 | SJTU ( <i>Topic: Transfer learning</i> )           |
| Tongkun Guan | 2019-2020 | HNU ( <i>Topic: Metal surface text detection</i> ) |
| Xunjin Wu    | 2018-2019 | SJTU ( <i>Topic: Transfer learning</i> )           |